Date: 15 February 2025 **Maximum Marks:** 5 Marks

Team ID: **Phase:** Project Planning Phase

PROJECT PLANNING & AGILE EXECUTION

Project: India's Agriculture Crop Production Analysis

This technical planning artifact outlines the product backlog structure, sprint breakdown schedules, team iteration metrics, and delivery burndown logics designed to iteratively build out the analytics platform pipeline.

1. Product Backlog & Sprint Allocation

SPRINT	EPIC / REQUIREMENT	STORY ID	USER STORY / CORE ENGINEERING TASK	POINTS	PRIORITY
Sprint-1	Data Ingestion & Pipelines	USN-01	As a system data analyst, I want an automated ETL script to ingest multi-state CSV crop yield sheets so that data is unified in a standardized schema.	5	HIGH
Sprint-1	Data Ingestion & Pipelines	USN-02	As a pipeline administrator, I want automated validation checks to tag missing or corrupted rainfall entries during ingestion.	3	HIGH
Sprint-1	Core Engine Backend	USN-03	As a backend system, I need database schemas partitioned by state and season (Kharif/Rabi) to keep query responses under 2.5 seconds.	5	HIGH
Sprint-2	Analytical Dashboard	USN-04	As a policy researcher, I want an interactive choropleth map dashboard to filter crop yield rates visually across India.	8	HIGH
Sprint-2	Analytical Dashboard	USN-05	As an agribusiness planner, I want to export filtered historical data subsets directly to structured formats like CSV/JSON.	3	MEDIUM
Sprint-3	Predictive Modelling Core	USN-06	As an analyst, I want to apply a predictive algorithm using past weather parameters to generate next-season crop production estimates.	8	HIGH

SPRINT	EPIC / REQUIREMENT	STORY ID	USER STORY / CORE ENGINEERING TASK	POINTS	PRIORITY
Sprint-3	Natural Language Interface	USN-07	As a field planner, I want an AI-assisted text search bar to query regional yields using simple language instead of SQL.	5	MEDIUM

2. Project Tracker & Release Roadmap

The following roadmap demonstrates the allocation of velocity metrics over uniform 6-day operating cycles to assure constant delivery outputs across the deployment window.

SPRINT	TARGET DELIVERABLE FOCUS	TOTAL STORY POINTS	DURATION	START DATE	PLANNED END
Sprint-1	Data Pipeline Ingestion & Database Architecture Core	20	6 Days	24 Oct 2022	29 Oct 2022
Sprint-2	Interactive Geospatial UI & Cross-Filtering Dashboard	20	6 Days	31 Oct 2022	05 Nov 2022
Sprint-3	Predictive Crop Yield ML Models & Natural Text Search	20	6 Days	07 Nov 2022	12 Nov 2022
Sprint-4	Infrastructure Hardening, Query Optimization & Deployment	20	6 Days	14 Nov 2022	19 Nov 2022

3. Agile Velocity Mechanics & Burn Down Logic

To assure rigorous delivery tracking, iteration speeds are formalized daily. The baseline calculations utilize the team velocity constraints to maintain clear timelines.

```
Target Velocity Per Sprint (V) = 20 Story Points
Sprint Fixed Duration (D) = 6 Days

Average Velocity per Iteration Unit (AV):
AV = Total Story Points / Sprint Duration = 20 / 6 = 3.33 Story Points / Day
```

Burn Down Configuration: Sprint-level validation uses an automated Burndown structure comparing absolute outstanding user story metrics on the vertical grid axis directly against day increments on the horizontal timeline. Deviations from the baseline slope trigger immediate mitigation evaluations during daily standalone synchronizations.